Tanmay Bhuskute

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SUMMARY

Software Engineer and M.S. Computer Science candidate (Dec 2026) with production experience designing, building, testing, and operating backend services and cloud infrastructure using Java, Python, Go, and TypeScript. Strong in REST API design, relational database schema and migrations, microservices, CI/CD testing, and observability.

EDUCATION

Master of Science in Computer Science Aug 2024 - Dec 2026
University of Cincinnati
Relevant Coursework: Advanced Algorithms, Database Theory, Intelligent Data Analysis.

Bachelor of Engineering in Computer Engineering Aug 2018 - Jul 2022
AISSMS College of Engineering
Relevant Coursework: Object-Oriented Programming, Data Analytics, Machine Learning.

TECHNICAL SKILLS

Programming: Java, Python, Go, TypeScript, JavaScript, C/C++, SQL, C#.

Backend & Data: Node.js, FastAPI, Flask, Express, REST APIs, GraphQL, Microservices, Message Queues (SQS); PostgreSQL, MySQL, Redis, schema design, indexing, query tuning, migrations.

Cloud & DevOps: AWS (EC2, S3, Lambda, SQS, RDS, CloudWatch, IAM), Docker, Kubernetes, Terraform, GitLab CI/CD, Jenkins, ArgoCD, Helm, Linux, SonarQube, Datadog.

Practices: Object-Oriented Design, SOLID, Design Patterns, TDD, Unit/Integration/E2E Testing, Code Review, Agile/Scrum.

WORK EXPERIENCE

Software Development Engineer | Siemens Industry Software Jul 2022 – Jun 2024

- Designed and maintained production backend automation for **Teamcenter X** in Python (Boto3) with object-oriented service modules and REST API integrations, replacing legacy Ruby scripts and cutting deployment time by **40%** across AWS environments.
- Migrated **40+ production environments** from process-based deployments to containerized **Kubernetes** microservices using ArgoCD, Helm, GitLab CI/CD, and HashiCorp Vault, an incremental legacy-modernization effort with backward-compatible cutover.
- Modeled environment config, provisioning state, and job status in **DynamoDB** and relational stores with indexed lookups, supporting idempotent retries and partial-failure recovery in CLI and Ansible workflows; added automated rollback cutting upgrade downtime **50%**.
- Executed **44 production environment upgrades** in 4 months, triaging live incidents across Linux, EC2, and databases using CloudWatch metrics, tracing, and Datadog structured-log correlation; as Quality Coach and **SonarQube** Specialist raised automated test coverage from **35% to 78%** via CI/CD quality gates and mentored **5 engineers**.

Graduate Researcher | University of Cincinnati Feb 2026 – Present

- Designed two-stage memory-augmented LLM pipeline for Android GUI bug reproduction as a config-driven, provider-agnostic service with a versioned run-artifact schema; evaluated against **ViBR** baseline across 53 runs on 20+ apps, showing **25.7x latency reduction** while preserving task-completion fidelity.

Software Developer Intern | Siemens Industry Software Mar 2022 – Jun 2022

- Automated AWS provisioning with **Terraform** infrastructure-as-code (**30%** faster deploys) and built a **Go + Terratest** framework catching **95%** of misconfigurations pre-production in CI.

Software Developer Intern | Government of Maharashtra, Water Resource Department Sep 2021 – Dec 2021

- Built and deployed a full-stack **React and MySQL** workflow application that digitized departmental record management and improved operational efficiency by **40%**.

PROJECTS AND RESEARCH

PolicyPilot – Policy Intelligence Portal [React, TS, Node, Express, Gemini, MCP] May 2026

- Built full-stack TypeScript app (React SPA + **Express 5 REST API** + tool server) on one Node.js process with **17 versioned tool APIs** and Zod schema validation at every data boundary; tested with **Vitest** suites in CI.

NYC Mobility Forecasting [Python, FastAPI, PostgreSQL, Redis, Docker] Jul 2026

- Built **FastAPI REST service** serving real-time demand predictions across **263 zones** from a **2.3M-row** dataset, backed by PostgreSQL with a **Redis** cache and containerized with Docker.

EXTRACURRICULAR ACTIVITIES

- **Finalist, RevolutionUC 2025 Hackathon** (300 participants, 31 schools) | **SAFe Practitioner 5.1** (Mar 2023).